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Assignment	01	Subject	AI

Question no 2

```
sentence = input("Enter a sentence: ")

words = sentence.split()

total_words = word(words)
total_characters = sent(sentence)
longest_word = long(words)
shortest_word = short(words)
total_vowels = vowel(sentence)

unique = set(words)
alphabetical = sorted(unique)

print("\nWORD ANALYSIS")

print("Sentence:", sentence)
print("Total Words:", total_words)
print("Total Characters:", total_characters)
print("Longest Word:", longest_word)
print("Shortest Word:", shortest_word)
print("Total Vowels:", total_vowels)
print("Unique Words:", unique)
print("Alphabetical Order:", alphabetical)

file = open("word_analysis.txt", "w")

file.write("WORD ANALYSIS\n")
file.write("==============\n")
file.write("Sentence: " + sentence + "\n")
file.write("Total Words: " + str(total_words) + "\n")
file.write("Total Characters: " + str(total_characters) + "\n")
file.write("Longest Word: " + longest_word + "\n")
file.write("Shortest Word: " + shortest_word + "\n")
file.write("Total Vowels: " + str(total_vowels) + "\n")
file.write("Unique Words: " + str(unique) + "\n")
file.write("Alphabetical Order: " + str(alphabetical) + "\n")
```

```
file.write("Total Vowels: " + str(total_vowels) + "\n")
file.write("Unique Words: " + str(unique) + "\n")
file.write("Alphabetical Order: " + str(alphabetical) + "\n")

file.close()

print("\nAnalysis saved in word_analysis.txt")
```

```
Enter a sentence:  my name is amna Aslam

WORD ANALYSIS
Sentence: my name is amna Aslam
Total Words: 5
Total Characters: 21
Longest Word: Aslam
Shortest Word: my
Total Vowels: 7
Unique Words: {'my', 'name', 'Aslam', 'amna', 'is'}
Alphabetical Order: ['Aslam', 'amna', 'is', 'my', 'name']

Analysis saved in word_analysis.txt
```

Question no 1

```
[1]: students = []

for i in range(5):
    name = input("Enter the name: ")
    roll = input("Enter the roll: ")
    ai = int(input("Enter the AI marks: "))
    python = int(input("Enter the Python marks: "))
    maths = int(input("Enter the Math marks: "))

    student = {
        "name": name,
        "roll_no": roll,
        "python": python,
        "ai": ai,
        "maths": maths
    }

    students.append(student)


for student in students:

    obtained = student["python"] + student["ai"] + student["maths"]
    percentage = (obtained / 300) * 100

    if percentage >= 80:
        grade = "A"
    elif percentage >= 70:
        grade = "B"
    elif percentage >= 60:
        grade = "C"
    elif percentage >= 50:
        grade = "D"
    else:
```

```
        grade = "D"
    else:
        grade = "F"

    student["obtained"] = obtained
    student["percentage"] = percentage
    student["grade"] = grade


print("\n=== STUDENT RESULTS ===")

for student in students:
    print("\nName:", student["name"])
    print("Roll Number:", student["roll_no"])
    print("Python:", student["python"])
    print("Artificial Intelligence:", student["ai"])
    print("Mathematics:", student["maths"])
    print("Obtained:", student["obtained"])
    print("Percentage:", student["percentage"], "%")
    print("Grade:", student["grade"])

highest = students[0]

for student in students:
    if student["percentage"] > highest["percentage"]:
        highest = student


print("\n========== HIGHEST PERCENTAGE ==========")
print("Name:", highest["name"])
print("Roll Number:", highest["roll_no"])
print("Percentage:", highest["percentage"], "%")
print("Grade:", highest["grade"])
```

```
with open("results.txt", "w") as file:

    file.write("========== STUDENT RESULTS ==========\n")

    for student in students:
        file.write("\nName: " + student["name"] + "\n")
        file.write("Roll Number: " + student["roll_no"] + "\n")
        file.write("Python: " + str(student["python"]) + "\n")
        file.write("Artificial Intelligence: " + str(student["ai"]) + "\n")
        file.write("Mathematics: " + str(student["maths"]) + "\n")
        file.write("Obtained: " + str(student["obtained"]) + "\n")
        file.write("Percentage: " + str(student["percentage"]) + "%\n")
        file.write("Grade: " + student["grade"] + "\n")

    file.write("\n========== HIGHEST PERCENTAGE ==========\n")
    file.write("Name: " + highest["name"] + "\n")
    file.write("Roll Number: " + highest["roll_no"] + "\n")
    file.write("Percentage: " + str(highest["percentage"]) + "%\n")
    file.write("Grade: " + highest["grade"] + "\n")

print("Results saved in results.txt")
```

```
Enter the name:  Amna Aslam
Enter the roll:  2
Enter the AI marks:  23
Enter the Python marks:  33
Enter the Math marks:  33
Enter the name:  alo
Enter the roll:  3
Enter the AI marks:  11
Enter the Python marks:  22
Enter the Math marks:  33

=== STUDENT RESULTS ===

Name: Amna Aslam
Roll Number: 2
Python: 33
Artificial Intelligence: 23
Mathematics: 33
Obtained: 89
Percentage: 29.666666666666668 %
Grade: F

Name: alo
Roll Number: 3
Python: 22
Artificial Intelligence: 11
Mathematics: 33
Obtained: 66
Percentage: 22.0 %
Grade: F

========== HIGHEST PERCENTAGE ==========
Name: Amna Aslam
Roll Number: 2
Percentage: 29.666666666666668 %
Grade: F
```

Question no 3

```
Saved in word_analysis.txt
```

[2]:

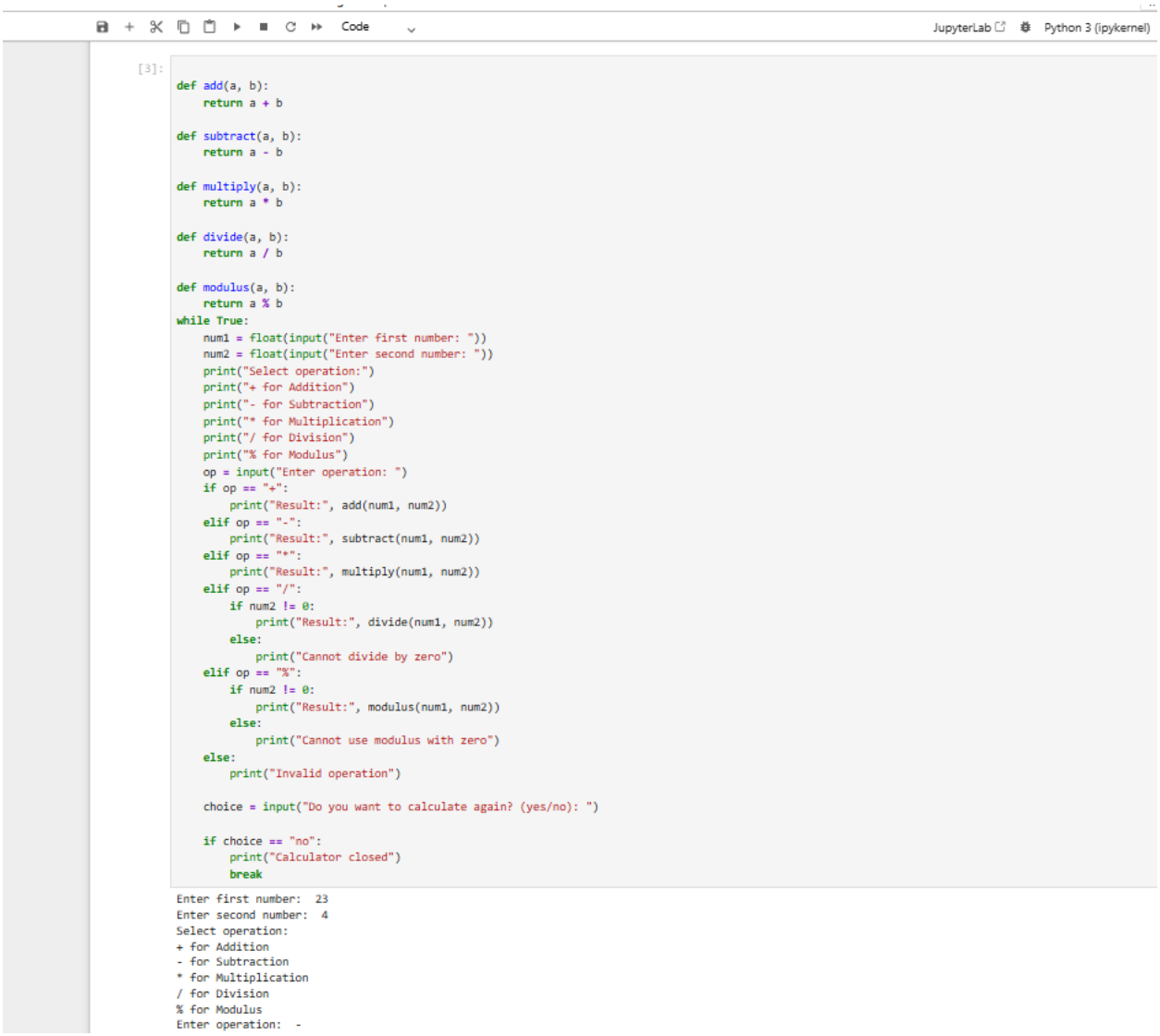
```
def bill(product):
    total = 0
    for p in product:
        amount = p["price"] * p["quantity"]
        total = total + amount
    return total
product = []
for i in range(5):
    p_name = input("Enter the product name: ")
    p_price = int(input("Enter the price: "))
    p_quantity = int(input("Enter the product quantity: "))
    data = {
        "name": p_name,
        "price": p_price,
        "quantity": p_quantity
    }

    product.append(data)
totel = bill(product)
print("\nShopping Bill")
for p in product:
    amount = p["price"] * p["quantity"]
    print(p["name"], "=", amount)
discount = 0
if totel > 10000:
    discount = totel * 10 / 100
final_bill = totel - discount
expensive = product[0]
for p in product:
    if p["price"] > expensive["price"]:
        expensive = p
print("Total Bill:", totel)
print("Discount:", discount)
print("Final Bill:", final_bill)
print("Most Expensive Product:", expensive["name"])
f = open("bill.txt", "w")

f.write("Shopping Bill\n")
for p in product:
    amount = p["price"] * p["quantity"]
    f.write(p["name"] + " = " + str(amount) + "\n")
f.write("Total Bill: " + str(totel) + "\n")
f.write("Discount: " + str(discount) + "\n")
f.write("Final Bill: " + str(final_bill) + "\n")
f.write("Most Expensive Product: " + expensive["name"] + "\n")
f.close()
print("Bill saved in bill.txt")
```

```
Enter the product name:  aa
Enter the price:  33
Enter the product quantity:  3
Enter the product name:  ss
Enter the price:  34
Enter the product quantity:  4
Enter the product name:  aff
Enter the price:  3
```

Question no 4



Question no 5

```
Calculator closed
```

[5]:

```python
def calculate_bill(order):
    total = 0

    for item in order:
        total = total + order[item]["price"] * order[item]["quantity"]

    return total


menu = {
    1: {"name": "Burger", "price": 500},
    2: {"name": "Pizza", "price": 800},
    3: {"name": "Fries", "price": 250},
    4: {"name": "Drink", "price": 150}
}

order = {}

while True:
    print("\n===== RESTAURANT MENU =====")
    print("1. Burger  Rs. 500")
    print("2. Pizza  Rs. 800")
    print("3. Fries  Rs. 250")
    print("4. Drink  Rs. 150")
    print("5. Finish Order")

    choice = int(input("Select item: "))

    if choice == 5:
        break

    elif choice >= 1 and choice <= 4:
        quantity = int(input("Enter quantity: "))

        if choice in order:
            order[choice]["quantity"] += quantity
        else:
            order[choice] = {
                "name": menu[choice]["name"],
                "price": menu[choice]["price"],
                "quantity": quantity
            }

        print("Item added to order!")

    else:
        print("Invalid choice!")


total = calculate_bill(order)

print("\n===== YOUR BILL =====")

for item in order:
    name = order[item]["name"]
    price = order[item]["price"]
```

```
for item in order:
    name = order[item]["name"]
    price = order[item]["price"]
    quantity = order[item]["quantity"]
    amount = price * quantity

    print(name, "x", quantity, "=", amount)

print("Final Bill: Rs.", total)


file = open("orders.txt", "w")

file.write("===== RESTAURANT BILL =====\n")

for item in order:
    name = order[item]["name"]
    price = order[item]["price"]
    quantity = order[item]["quantity"]
    amount = price * quantity

    file.write(name + " x " + str(quantity) + " = Rs. " + str(amount) + "\n")

file.write("Final Bill: Rs. " + str(total) + "\n")

file.close()

print("Bill saved in orders.txt")
```

```
===== RESTAURANT MENU =====
1. Burger  Rs. 500
2. Pizza  Rs. 800
3. Fries  Rs. 250
4. Drink  Rs. 150
5. Finish Order
Select item:  3
Enter quantity:  2
Item added to order!

===== RESTAURANT MENU =====
1. Burger  Rs. 500
2. Pizza  Rs. 800
3. Fries  Rs. 250
4. Drink  Rs. 150
5. Finish Order
Select item:  2
Enter quantity:  33
Item added to order!

===== RESTAURANT MENU =====
1. Burger  Rs. 500
2. Pizza  Rs. 800
3. Fries  Rs. 250
4. Drink  Rs. 150
5. Finish Order
Select item:  2
Enter quantity:  1
Select item:  5

===== YOUR BILL =====
Fries x 5 = 1250
Pizza x 39 = 31200
Final Bill: Rs. 32450
Bill saved in orders.txt
```

[ ]: